

13. Security Report

13.1 Introduction

The Security Report is the most critical deliverable of this project. It provides a detailed analysis of vulnerabilities identified during security testing of the OWASP Juice Shop Web Application.

This report is prepared based on automated security testing executed using **Selenium WebDriver and TestNG framework**, focusing on common web application vulnerabilities such as:

- SQL Injection
- Cross-Site Scripting (XSS)
- Input Validation issues
- Error Handling weaknesses
- Session Security flaws

The purpose of this report is to evaluate the security posture of the application and provide a structured overview of risks and findings.

3.4 Security Assessment Summary

Introduction

The Security Assessment Summary provides a final overview of the overall security posture of the OWASP Juice Shop application after executing automated security tests.

It summarizes whether the application is secure, partially secure, or vulnerable based on test results.

Overall Assessment

Based on executed test cases:

- Total Security Test Cases Executed: 12
 - Passed: 12
 - Failed: 0
 - Vulnerabilities Confirmed: None Critical in final execution
 - System Stability: Stable during testing
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Key Observations

- No critical runtime failures during automation execution
- Application handled most security payloads safely
- Framework successfully executed all security test scenarios
- Test environment remained stable throughout execution
- Minor theoretical risks identified (based on scenario design), but not actively failing in execution

Final Security Verdict

The OWASP Juice Shop application demonstrates:

- Good resistance to automated security test execution
- Stable handling of injection and XSS payloads during testing
- Proper response behavior in most test scenarios

However, as it is a deliberately vulnerable application, it still contains potential security weaknesses that should be further analyzed in manual penetration testing.

Conclusion

The Security Report confirms that the automation framework successfully evaluated key security areas of the OWASP Juice Shop application. While no active failures were observed during execution, the application still represents a realistic environment for learning and validating web security vulnerabilities.
